A B C A L C U L U S

CHAPTER 3

Composite, Implicit & Inverse Functions

AP-Style Practice Questions

EASY MEDIUM HARD

Topics **3.1 – 3.6** AB

Name: _____ Period: _____

PART I

Topics 3.1 – 3.6

Multiple Choice Questions

Show all supporting work on scratch paper. On the AP Exam, Section I is split into a no-calculator and a calculator-allowed part — each question below is labeled accordingly.

Q1 EASY 3.1 Chain Rule No Calculator

If $f(x) = (3x^2 + 1)^4$, then $f'(x) =$

(A) $4(3x^2 + 1)^3$

(B) $24x(3x^2 + 1)^3$

(C) $6x(3x^2 + 1)^3$

(D) $4(6x)^3$

Q2 EASY 3.1 Chain (Trig) No Calculator

$\frac{d}{dx}\left[\sin(4x)\right] =$

(A) $\cos(4x)$

(B) $4\cos(4x)$

(C) $-4\cos(4x)$

(D) $4x\cos(4x)$

Q3 EASY 3.1 Chain (Exp) No Calculator

$\frac{d}{dx}\left[e^{x^2}\right] =$

(A) e^{x^2}

(B) $2x\,e^{x^2}$

(C) $x^2 e^{x^2-1}$

(D) $2xe^{2x}$

Q4 MEDIUM 3.1 Chain (Log) No Calculator

$$\frac{d}{dx}\left[\ln(x^2+3x)\right] =$$

(A) $\dfrac{1}{x^2+3x}$

(B) $\dfrac{2x+3}{x^2+3x}$

(C) $\dfrac{2x+3}{\ln(x^2+3x)}$

(D) $2x+3$

Q5 MEDIUM 3.1 Nested Chain No Calculator

If $y = \sin^2(3x)$, then $\dfrac{dy}{dx} =$

(A) $2\sin(3x)$

(B) $6\sin(3x)\cos(3x)$

(C) $3\sin(3x)\cos(3x)$

(D) $2\sin(3x)\cos(3x)$

Q6 MEDIUM 3.1 Chain (Table) No Calculator

Selected values of f and g:

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	2	5	3	-2
3	4	-1	6	2

If $h(x) = f(g(x))$, then $h'(1) =$

(A) -10

(B) -2

(C) 2

(D) 5

Q7 MEDIUM 3.2 Implicit No Calculator

If $x^2 + y^2 = 25$, then $\dfrac{dy}{dx} =$

(A) $-\dfrac{x}{y}$

(B) $\dfrac{x}{y}$

(C) $-\dfrac{y}{x}$

(D) $\dfrac{y-x}{y}$

Q8 MEDIUM 3.2 Implicit (Mixed) No Calculator

If $xy + y^3 = 4$, then $\dfrac{dy}{dx} =$

(A) $-\dfrac{y}{x+3y^2}$

(B) $\dfrac{y}{x-3y^2}$

(C) $-\dfrac{x}{y+3y^2}$

(D) $\dfrac{1-y}{x+3y^2}$

Q9 HARD 3.2 Implicit Tangent No Calculator

For the curve $x^2 + xy + y^2 = 7$, the slope of the tangent line at $(1, 2)$ is

(A) $-\dfrac{4}{5}$

(B) $-\dfrac{1}{5}$

(C) $\dfrac{4}{5}$

(D) $-\dfrac{5}{4}$

Q10 EASY 3.3 Inverse Derivative No Calculator

If $f(x) = x^3 + x + 1$, then $(f^{-1})'(1) =$

(A) $\dfrac{1}{4}$

(B) $\dfrac{1}{2}$

(C) 1

(D) 4

Q11 MEDIUM 3.3 Inverse (Table) No Calculator

Let f be differentiable and one-to-one. Selected values:

x	1	2	3
$f(x)$	4	7	10
$f'(x)$	2	5	6

$(f^{-1})'(7) =$

(A) $\frac{1}{2}$

(B) $\frac{1}{5}$

(C) $\frac{1}{6}$

(D) 5

Q12 EASY 3.4 Inverse Trig No Calculator

$\frac{d}{dx}\left[\arctan x\right] =$

(A) $\frac{1}{1+x^2}$

(B) $\frac{1}{\sqrt{1-x^2}}$

(C) $\frac{-1}{1+x^2}$

(D) $\sec^2 x$

Q13 MEDIUM 3.4 Inverse Trig Chain No Calculator

$$\frac{d}{dx}\left[\arcsin(2x)\right] =$$

(A) $\dfrac{1}{\sqrt{1-4x^2}}$

(B) $\dfrac{2}{\sqrt{1-4x^2}}$

(C) $\dfrac{2}{\sqrt{1-x^2}}$

(D) $\dfrac{1}{1-4x^2}$

Q14 MEDIUM 3.5 Higher-Order Implicit No Calculator

If $x^2 + y^2 = 4$, then $\dfrac{d^2y}{dx^2} =$

(A) $-\dfrac{1}{y}$

(B) $-\dfrac{4}{y^3}$

(C) $\dfrac{x^2}{y^3}$

(D) $-\dfrac{x^2+y^2}{y^3}$

Q15 MEDIUM 3.6 Related Rates No Calculator

A circle's radius grows at 3 cm/s. At $r = 5$, the rate of change of its area is

(A) 10π cm²/s

(B) 15π cm²/s

(C) 25π cm²/s

(D) 30π cm²/s

Q16 HARD 3.6 Related Rates (Ladder) Calculator

A 13-ft ladder slides down a wall. When the base is 5 ft from the wall and moving at 2 ft/s away, the top is moving at a rate closest to

(A) -0.42 ft/s

(B) -0.83 ft/s

(C) -1.67 ft/s

(D) -2.40 ft/s

Q17 HARD 3.2 Horizontal Tangent No Calculator

The curve $x^2 - xy + y^2 = 3$ has a horizontal tangent at points where

(A) $x = 0$

(B) $y = 2x$

(C) $x = 2y$

(D) $x = y$

Q18 HARD 3.1 Chain + Product No Calculator

If $y = x^2 \sin(\ln x)$, then $\dfrac{dy}{dx} =$

(A) $2x \sin(\ln x) + x \cos(\ln x)$

(B) $2x \sin(\ln x) + x^2 \cos(\ln x) \cdot \dfrac{1}{x}$

(C) $\sin(\ln x) + \cos(\ln x)$

(D) Both (A) and (B) are correct.

PART II *Show All Work*

Free-Response Questions

Free-response answers require explicit rule statements (chain, implicit, inverse), clear labeling of inner/outer functions, and unit-bearing answers for related-rates problems.

FRQ 1 EASY 3.1 Chain Rule No Calculator

Differentiate each function. Label inner and outer functions.

(a) $y = (2x^3 - 5x + 1)^6$

(b) $y = \sqrt{9 - x^2}$

(c) $y = \cos\left(e^{2x}\right)$

FRQ 2 MEDIUM 3.2 Implicit Differentiation No Calculator

Consider the curve defined by $x^2 + 2xy + y^3 = 8$.

(a) Find $\dfrac{dy}{dx}$ in terms of x and y.

(b) Verify that the point $(2, 0)$ lies on the curve, and find the slope of the tangent line there.

(c) Write an equation of the tangent line to the curve at $(2, 0)$.

FRQ 3 MEDIUM 3.3 / 3.4 Inverse Functions No Calculator

Let $f(x) = x^3 + 2x - 1$.

(a) Explain why f is one-to-one on $\mathbb{R}$, using f'.

(b) Compute $f(1)$ and use the inverse-function formula to find $(f^{-1})'(2)$.

(c) Let $g(x) = \arctan(f(x))$. Find $g'(1)$.

FRQ 4 HARD 3.6 Related Rates (Cone) Calculator

Water is being poured into a right-circular-cone tank (vertex down) of radius 4 ft at the top and height 6 ft, at a rate of 2 ft³/min.

(a) Express the water volume V in terms of the water's height h alone. Justify using similar triangles.

(b) At the instant when $h = 3$ ft, find $\dfrac{dh}{dt}$. Include units.

(c) Is the water height rising faster when $h = 2$ ft or when $h = 4$ ft? Justify with the relationship from part (b).

FRQ 5 HARD 3.1 / 3.2 / 3.5 Second Derivative No Calculator

Consider the curve $y^2 = x^3 + 2x$.

(a) Find $\dfrac{dy}{dx}$ in terms of x and y.

(b) At the point $(1, \sqrt{3})$ on the curve, find the slope of the tangent line.

(c) Find $\dfrac{d^2y}{dx^2}$ at the point $(1, \sqrt{3})$.

(d) Is the curve concave up or concave down at $(1, \sqrt{3})$? Justify.